Faculty of Engineering
Stellenbosch University
Fakulteit Ingenieurswese
Universiteit Stellenbosch

ENGINEERING
EZOBUNJINELI
INGENIEURSWESE

Electronics 245 **Elektronika 245**	**Assessment 2** November 2020 **Assessering 2** November 2020

Duration **Tydsduur**	**3 h**	**Full marks** **Volpunte**	**90**

	Names ***Name***	**Signature** ***Handtekening***
Examiner(s) *Eksaminator(e)*	Dr. M. Groch	
Moderator(s) *Moderator(e)*	Prof. M. J. Kamper	

Rules and Instructions	**Reëls en Voorskrifte**
The marks for each question are indicated to the right of the question in square brackets.	Die punte vir elke vraag word regs van die vraag in vierkantige hakies aangedui.
Pocket calculators prescribed for the first two years of the BEng may be used.	**Sakrekenaars** voorgeskryf vir die eerste twee jaar van die BIng mag gebruik word.
This is an **open-book assessment, but** unless the question paper expressly states otherwise, you may **consult only the following sources** from the time that the assessment becomes available on SUNLearn until you have submitted all your answers: the textbook and sources prescribed, recommended or provided by the lecturers; notes or materials that you created personally and independently using the aforementioned sources.	Hierdie is 'n **oopboek-assessering**, **maar** tensy die vraestel uitdruklik iets anders stel, mag jy **slegs die volgende inligtingsbronne raadpleeg** vanaf die tyd dat die assessering op SUNLearn beskikbaar raak totdat jy al jou antwoorde ingedien het: die handboek en bronne wat die dosente voorgeskryf, aanbeveel of voorsien het; aantekeninge of materiaal wat jy persoonlik en onafhanklik vanaf die voorgenoemde bronne geskep het.
Communication with other candidates or with anybody else (except for the lecturer) is not allowed from the time the assessment becomes available on SUNLearn until you have finally submitted all your answers.	Kommunikasie met ander kandidate of met enigiemand anders (behalwe vir die dosent) word nie toegelaat nie vandat die assessering op SUNLearn beskikbaar gestel is totdat jy al jou antwoorde finaal ingedien het.

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NB: Answers must be handwritten on A4 sheets using dark pen, unless the question paper expressly instructs differently. Answers may not be typed on a computer or otherwise. **NB: The SUNLearn submission deadline is 30 minutes in addition to the normal writing time, plus extra writing time for students that qualify for it.** Only if you have difficulty submitting the assignment on SUNLearn before the deadline, you may email your submission to submitENG@sun.ac.za. Your email's subject line and message body must give your student number, the module name and the assessment (e.g. 12345678; Material Science A 244; A1). The body must further give an explanation of the nature of the difficulty encountered when submitting to SUNLearn. Appropriate evidence of the difficulty must be attached.	**NB: Antwoorde moet handgeskrewe op A4 blaaie met donker pen wees, tensy die vraestel uitdruklik anders voorskryf. Antwoorde mag nie op 'n rekenaar, of andersins, getik word nie.** **NB: Die SUNLearn spertyd vir indiening is 30 minute ekstra na normale skryftyd, plus ekstra skryftyd vergunning vir studente wat daarvoor kwalifiseer.** Slegs indien jy probleme ondervind met die indiening van jou werk op SUNLearn, mag jy jou werk epos na submitENG@sun.ac.za. Jou epos se onderwerp en die inhoud moet die volgende bevat: jou studentenommer, die modulenaam en die assessering (bv. 12345678; Materiaalkunde A 244; A1). Die inhoud moet ook 'n verduideliking gee van die aard van die probleme ondervind toe jy probeer indien het. Gepaste bewyse van die probleme moet aangeheg word.
NB: The following must be submitted separately on the relevant SUNLearn assignments: Question 1, Section A, Section B and Section C. When the answers for any section in the question paper have been completed, they should be scanned (e.g. using a cell phone app; "Genius Scan" and "Adobe Scan" recommended) and submitted as pdf file on the corresponding assignment on SUNLearn. It must be submitted without delay, to minimise the risk of not being able to submit answers. The cover page (question 1) must be submitted. You are allowed to resubmit answers and can therefore submit corrections after a previous version was submitted.	**NB: Die volgende moet afsonderlik op die betrokke SUNLearn "assignment" ingegee word: Vraag 1, Afdeling A, Afdeling B en Afdeling C.** Wanneer enige antwoord(e) vir enige vraag/afdeling van die vraestel voltooi is, moet dit sonder vertraging geskandeer word (bv deur 'n selfoon toep te gebruik; "Genius Scan" en "Adobe Scan" word aanbeveel) en as 'n pdf lêer op SUNLearn ingedien word, om die risiko dat antwoorde nie ingedien kan word nie, te minimeer. Die dekblad (vraag 1) moet ingedien word. Jy word toegelaat om 'n antwoord meer as een keer in te dien en kan dus korreksies indien nadat 'n vorige weergawe ingedien is.

Question 1 [0]	**Vraag 1 [0]**
Before completing this assessment, submit one cover page containing the following: • A copy of your student card (if not available, then ID document or passport). • Your student number, surname and initials written by hand. • The following declaration, unless a similar declaration was made on SUNLearn before starting the assignment: "I declare hereby that I have read the rules and instructions on the question paper and have abided by all the rules and instructions". • Your handwritten signature.	Voordat hierdie assessering voltooi word, gee in een dekblad wat die volgende bevat: • 'n Kopie van jou studentekaart (indien nie beskikbaar nie, dan ID dokument of paspoort). • Jou studentenommer, van en voorletters. Hierdie moet handgeskrewe wees. • Die volgende verklaring, tensy 'n soortgelyke verklaring gemaak is op SUNLearn voordat die "assignment" begin is: "Ek verklaar hiermee dat ek die reëls en instruksies op die vraestel gelees het en voldoen het aan al die reëls en instruksies". • Jou handgeskrewe handtekening.

NB: Read all the above instructions before starting to answer the remainder of the question paper.	NB: Lees al die bostaande instruksies voordat die res van die vraestel beantwoord word.
SECTION A	**AFDELING A**

Question 2 [20]

a) Explain the following terms **in your own words**:

i) Extrinsic semiconductor. (2)

ii) Majority charge carrier. (2)

iii) Built-in potential. (3)

iv) Drift current. (2)

b) A silicon semiconductor material is to be designed such that the majority carrier hole concentration is $p_o = 5 \text{ x } 10^{16} \text{ cm}^{-3}$. Answer the questions that follow.

i) Should donor or acceptor impurities be added to achieve this hole concentration? What concentration of dopant impurity atoms should be added? (2)

ii) What is the concentration of free electrons at $T = 324$ K? (4)

c) Determine the increase in forward bias voltage required to increase the conduction current by a factor of 20 for a silicon pn junction diode at a temperature of 315 K. Assume the diode is strongly biased in the forward direction. (5)

Vraag 2 [20]

a) Verduidelik die volgende terme **in jou eie woorde**:

i) Ekstrinsieke halfgeleier. (2)

ii) Meerderheidsladingsdraer. (2)

iii) Sperpotensiaal. (3)

iv) Dryfstroom. (2)

b) 'n Silikon halfgeleier materiaal moet ontwerp word sodat die holte-konsentrasie $p_o = 5 \text{ x } 10^{16} \text{ cm}^{-3}$ is. Beantwoord die vrae wat volg.

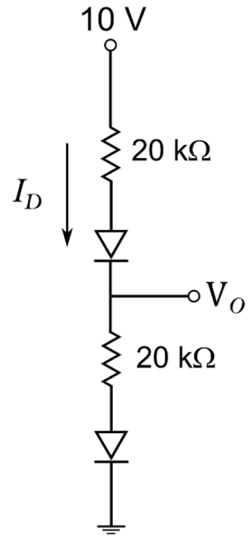
i) Moet donor- of akseptor onsuiwerhede bygevoeg word om hierdie holte konsentrasie te bereik? Watter konsentrasie van onsuiwerheid atome moet bygevoeg word? (2)

ii) Wat is die konsentrasie van vrye elektrone by $T = 324$ K? (4)

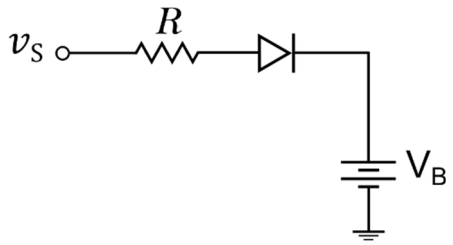
c) Bepaal die toename in voorwaartse voorspanning wat benodig word om die geleidings stroom met 'n faktor van 20 te verhoog vir 'n silikon pn voegvlak diode by 'n temperatuur van 315 K. Neem aan dat die diode sterk in die voorwaartse rigting voorgespan is. (5)

Question 3 [20]

a) Consider the circuit below. Calculate I_D and V_O if $V_\gamma = 0.7$ V and $r_f = 0\ \Omega$ for both diodes. (5)



b) A battery charging circuit is shown below. Assume that $V_B = 10$ V, $R = 200\ \Omega$, and $V_\gamma = 0.7$ V, $r_f = 0\ \Omega$. The input signal is given by $v_S = 25\sin(\omega t)$ V. Calculate the portion (percentage) of the total cycle of the input signal over which the diode is conducting current. (5)



c) A SPICE analysis is required for the circuit shown below. The component values are given by $V_1 = 20$ V, $R_1 = 2$ kΩ, $R_2 = 1$ MΩ, and $C_L = 15$ µF.

Answer the questions that follow.

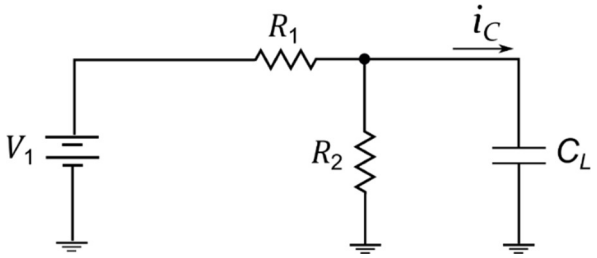
Vraag 3 [20]

a) Beskou die stroombaan hieronder. Bereken I_D en V_O as $V_\gamma = 0.7$ V en $r_f = 0\ \Omega$ vir beide diodes. (5)

b) 'n Batterylaaier word in die skets getoon. Aanvaar dat $V_B = 10$ V, $R = 200\ \Omega$, en $V_\gamma = 0.7$ V, $r_f = 0\ \Omega$. Die intreesein word gegee deur $v_S = 25\sin(\omega t)$ V. Bereken die breukdeel (persentasie) van die totale siklus van die intreesein wat die diode stroom gelei. (5)

c) 'n SPICE-ontleding is nodig vir die stroombaan hieronder Die komponent-waardes word gegee deur $V_1 = 20$ V, $R_1 = 2$ kΩ, $R_2 = 1$ MΩ, en $C_L = 15$ µF.

Beantwoord die vrae wat volg.



i) Redraw the circuit with node labels for analysis in NgSpice. Make provision in your circuit for the measurement of the current i_C. (2)

ii) Give the NgSpice code that will correctly define the components in the circuit. Component values are given above. (6)

iii) Give the line of code to plot the voltage over the resistor R_1, and the line of code to plot the current i_C. (2)

i) Teken die stroombaan oor met 'n naam by elke nodus sodat dit in NgSpice geanaliseer kan word. Maak in u stroombaan voorsiening vir die meet van die stroom i_C. (2)

ii) Gee die NgSpice-kode wat die komponente in die stroombaan korrek sal definieer. Komponent-waardes word hierbo gegee. (6)

iii) Gee die lyn van kode om die spanning oor die weerstand, R_1, te plot, en die lyn van kode om die stroom i_C te plot. (2)

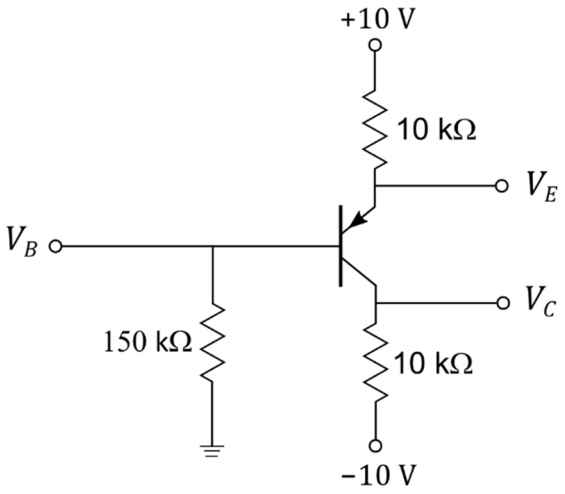
SECTION B	**AFDELING B**

Question 4 [25]

a) In the circuit shown, measurement indicates $V_{EB(on)}$ to be +0.7 V, and V_E to be +1.7 V. Calculate the values of α and β for this transistor. State and prove all assumptions. (8)

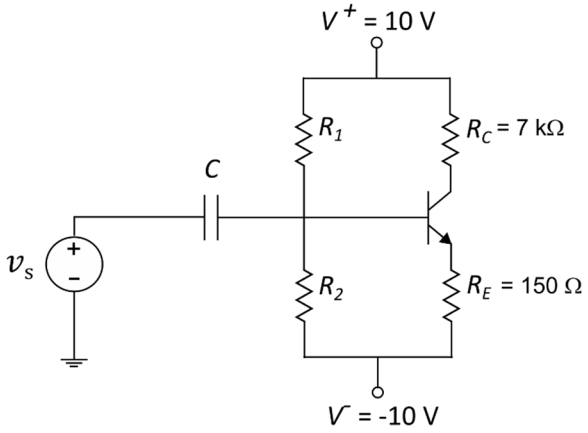
Vraag 4 [25]

a) Metings in die getoonde stroombaan dui aan dat $V_{EB(on)}$ gelyk aan +0.7 V is, en V_E gelyk aan +1.7 V is. Bereken die waardes van α en β vir hierdie transistor. Stel en bewys alle aannames. (8)



b) Consider the circuit below. Assume transistor parameters of $\beta = 100$, $V_A \rightarrow \infty$ and $V_{BE(on)} = 0.7\,\text{V}$. Answer the questions that follow.

b) Beskou die stroombaan hieronder. Neem aan transistor parameters van $\beta = 100$, $V_A \rightarrow \infty$ en $V_{BE(on)} = 0.7\,\text{V}$. Beantwoord die vrae wat volg.



i) Sketch the load line and select V_{CEQ} for optimal output swing. Do not consider saturation when selecting the value. (2)

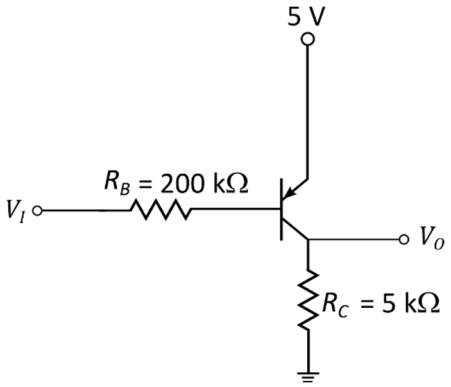
i) Skets die laslyn en kies V_{CEQ} vir optimale uittreeswaai. Moet nie versadiging in ag neem wanneer u die waarde kies nie. (2)

ii) Design a bias stable amplifier using the selected Q-point from (i). (7)

ii) Ontwerp 'n versterker met werkpuntstabiliteit. Maak gebruik van die Q-punt uit (i). (7)

c) A transistor circuit is shown in the sketch below. The transistor parameters are given as $V_{EB(on)} = 0.7\,\text{V}$, $\beta = 100$, $V_{EC(sat)} = 0.2\,\text{V}$ and $V_A \rightarrow \infty$. Draw the transfer curve (V_I vs V_O) of the circuit when the input voltage is varied from 0 to 5 V. Show the important values and label each operating mode on the graph. (8)

c) 'n Transistorbaan word in die skets getoon hieronder. Die transistor parameters word gegee as $V_{EB(on)} = 0.7\,\text{V}$, $\beta = 100$, $V_{EC(sat)} = 0.2\,\text{V}$ and $V_A \rightarrow \infty$. Teken die oordrag kurwe (V_I vs V_O) van die stroombaan wanneer die intreespanning van 0 tot 5 V verander word. Toon die belangrike waardes en benoem elke modus van operasie op die grafiek. (8)



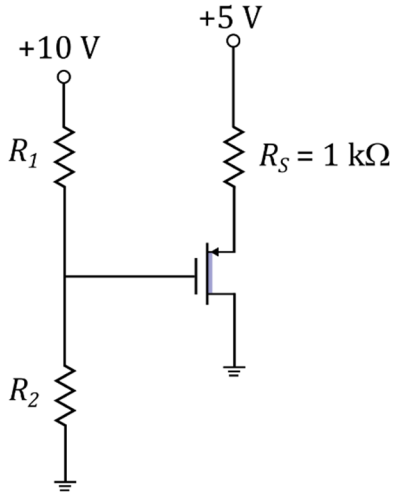
SECTION C	AFDELING C

Question 5 [25]

a) Design the circuit below so that $V_{SD} = 3$ V, and the current in the bias resistors is 10 % of the drain current. The transistor parameters are $V_{TP} = +1.5$ V and $K_p = 0.5$ mA/V^2. (10)

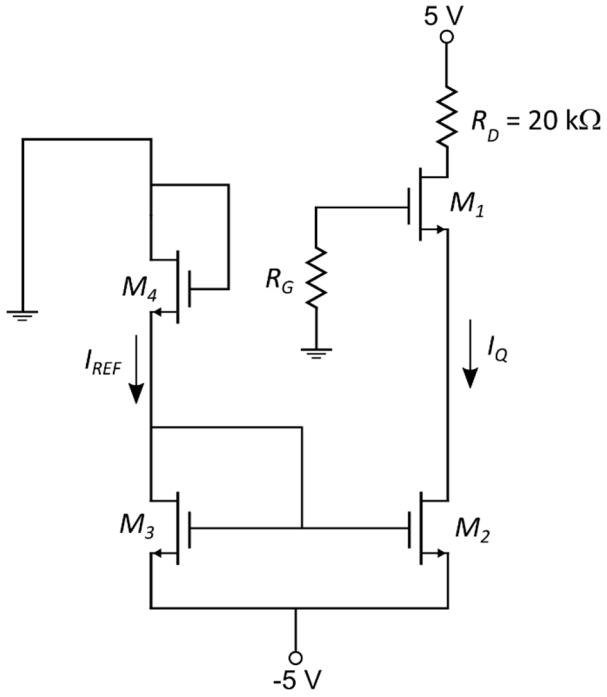
Vraag 5 [25]

a) Ontwerp die stroombaan hieronder sodat $V_{SD} = 3$ V, en die stroom in die voorspanningweerstande 10 % van die afvoerstroom is. Die transistorparameters is $V_{TP} = +1.5$ V en $K_p = 0.5$ mA/V^2. (10)



b) Consider the diagram below. The parameters $V_{TN} = 1.2$ V, and $k_n' = 60$ µA/V^2 for all transistors. Assume $V_{GS2} = 3.2$ V, $V_{GS1} = 2$ V and $I_Q = 150$ µA. Answer the questions that follow.

b) Beskou die onderstaande diagram. Die parameters $V_{TN} = 1.2$ V, en $k_n' = 60$ µA/V^2 vir alle transistors. Aanvaar $V_{GS2} = 3.2$ V, $V_{GS1} = 2$ V en $I_Q = 150$ µA. Beantwoord die vrae wat volg.



i) Determine the ratio of K_{n4}/K_{n3}. (5)

ii) Prove that both M_1 and M_2 are biased in saturation mode. (7)

iii) Determine the width-to-length ratio for M_2. (3)

i) Bepaal die verhouding K_{n4}/K_{n3}. (5)

ii) Bewys dat beide M_1 en M_2 in die versadigings modus voorgespan is. (7)

iii) Bepaal die breedte-tot-lengte verhouding vir M_2. (3)

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$$n_i = BT^{3/2}e^{-E_g/2kT}$$
$$n_o p_o = n_i^2$$
$$J_n = en\mu_n E$$
$$J_p = ep\mu_p E$$
$$\sigma = e\mu_n n + e\mu_p p$$
$$J = \sigma E = \frac{1}{\rho}E$$
$$J_n = eD_n\frac{dn}{dx}$$
$$J_p = -eD_p\frac{dp}{dx}$$
$$\frac{D_p}{\mu_p} = \frac{D_n}{\mu_n} = \frac{kT}{e} = V_T$$

$k = 8,62 \times 10^{-5}$ eV/K = 1,38 x 10^{-23} J/K

$e = 1,6 \times 10^{-19}$ C

$V_T = 26$ mV (300K)

$$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$
$$i_D = I_S\left(e^{v_D/nV_T} - 1\right)$$
$$\frac{1}{r_D} = \frac{I_{DQ}}{V_T}$$
$$V_r = \frac{V_M}{2fRC} \text{ (Full-wave)}$$
$$i_{D\text{peak}} = \begin{cases} \frac{V_M}{R}\left(1 + \pi\sqrt{\frac{2V_M}{V_r}}\right) & \text{(Full)} \\ \frac{V_M}{R}\left(1 + 2\pi\sqrt{\frac{2V_M}{V_r}}\right) & \text{(Half)} \end{cases}$$
$$i_{D\text{avg}} = \begin{cases} \frac{1}{\pi}\sqrt{\frac{2V_r}{V_M}}.\frac{V_M}{R}\left(1 + \frac{\pi}{2}\sqrt{\frac{2V_M}{V_r}}\right) & \text{(Full)} \\ \frac{1}{2\pi}\sqrt{\frac{2V_r}{V_M}}.\frac{V_M}{R}\left(1 + \pi\sqrt{\frac{2V_M}{V_r}}\right) & \text{(Half)} \end{cases}$$
$$v_{avg} = \frac{1}{T}\int_0^T v(t)dt$$
$$\text{Source reg.} = \frac{\Delta v_L}{\Delta v_{PS}} \times 100\%$$
$$\text{Load reg.} = \frac{v_{L(\text{no load})} - v_{L(\text{full load})}}{v_{L(\text{full load})}} \times 100\%$$

$$i_E = I_{EO}e^{v_{BE}/V_T}$$
$$i_C = I_S e^{v_{BE}/V_T}\left(1 + \frac{v_{CE}}{V_A}\right)$$
$$i_C = \alpha i_E = \beta i_B$$
$$\frac{1}{r_o} = \left.\frac{\partial i_C}{\partial v_{CE}}\right|_{V_{BE}} = \frac{I_C}{V_A}$$
$$BV_{CEO} = \frac{BV_{CBO}}{\sqrt[n]{\beta}}$$
$$g_m = \frac{I_{CQ}}{V_T}$$
$$r_\pi = \frac{\beta}{g_m} = \frac{\beta V_T}{I_{CQ}} = \frac{V_T}{I_{BQ}}$$
$$R_{TH} = 0.1\beta R_E$$
$$K_n = \frac{k_n'}{2}\left(\frac{W}{L}\right)$$
$$k_n' = \mu_n C_{ox}$$
$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$
$$i_D = K_n\left(v_{GS} - V_{TN}\right)^2\left(1 + \lambda v_{DS}\right)$$
$$i_D = K_n[2(v_{GS} - V_{TN})v_{DS} - v_{DS}^2]$$
$$v_{DS(\text{sat})} = v_{GS} - V_{TN}$$
$$i_D = K_p\left(v_{SG} + V_{TP}\right)^2\left(1 + \lambda v_{SD}\right)$$
$$i_D = K_p[2(v_{SG} + V_{TP})v_{SD} - v_{SD}^2]$$
$$v_{SD(\text{sat})} = v_{SG} + V_{TP}$$
$$\frac{1}{r_o} = \left.\frac{\partial i_D}{\partial v_{DS}}\right|_{V_{GSQ}} = \lambda I_{DQ}$$

III	IV	V
^{5}B		
^{13}Al	^{14}Si	^{15}P
^{31}Ga	^{32}Ge	^{33}As
^{49}In		

Suffix	Name	Factor
T	Tera	10^{12}
G	Giga	10^{9}
Meg	Mega	10^{6}
K	Kilo	10^{3}
mil	Mil	25.4×10^{-6}
m	milli	10^{-3}
u	micro	10^{-6}
n	nano	10^{-9}
p	pico	10^{-12}
f	femto	10^{-15}

Table 2.2: Ngspice scale factors

Material	*Eg* (eV)	*B* (cm^{-3} K$^{-3/2}$)
Silicon (Si)	1.1	5.23×10^{15}
Gallium arsenide (GaAs)	1.4	2.10×10^{14}
Germanium (Ge)	0.66	1.66×10^{15}